

Ph4nt0m 1ntrud3r

picoCTF > Forensics · + Tags

I have to analyze the packet capture file in Wireshark to find the flag. Additionally, the hint was to filter data by time. So, I sort the packets by time.

No.	Time	Source	Destination	Protocol	Length	Info
2	0.002604	192.168.0.2	192.168.1.2	TCP	48	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=8
3	0.002141	192.168.0.2	192.168.1.2	TCP	48	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=8
4	0.001899	192.168.0.2	192.168.1.2	TCP	48	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=8
22	0.001560	192.168.0.2	192.168.1.2	TCP	48	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=8
13	0.001298	192.168.0.2	192.168.1.2	TCP	48	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=8
9	0.001034	192.168.0.2	192.168.1.2	TCP	48	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=8
15	0.000720	192.168.0.2	192.168.1.2	TCP	48	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=8
16	0.000476	192.168.0.2	192.168.1.2	TCP	48	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=8
21	0.000228	192.168.0.2	192.168.1.2	TCP	48	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=8
1	0.000000	192.168.0.2	192.168.1.2	TCP	48	20 → 80 [SYN] Seq=0 Win=8192 Len=8 [TCP PDU reassembled in 6]
18	0.000229	192.168.0.2	192.168.1.2	TCP	48	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=8
7	0.000453	192.168.0.2	192.168.1.2	TCP	48	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=8
8	0.000676	192.168.0.2	192.168.1.2	TCP	48	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=8
5	0.000935	192.168.0.2	192.168.1.2	TCP	48	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=8
10	0.001165	192.168.0.2	192.168.1.2	TCP	48	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=8
17	0.001388	192.168.0.2	192.168.1.2	TCP	52	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=12
6	0.001804	192.168.0.2	192.168.1.2	TCP	52	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=12 [TCP PDU reassembled in 6]
20	0.002045	192.168.0.2	192.168.1.2	TCP	52	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=12
14	0.002367	192.168.0.2	192.168.1.2	TCP	52	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=12
19	0.002586	192.168.0.2	192.168.1.2	TCP	52	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=12
11	0.002819	192.168.0.2	192.168.1.2	TCP	52	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=12
12	0.003036	192.168.0.2	192.168.1.2	TCP	44	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=4

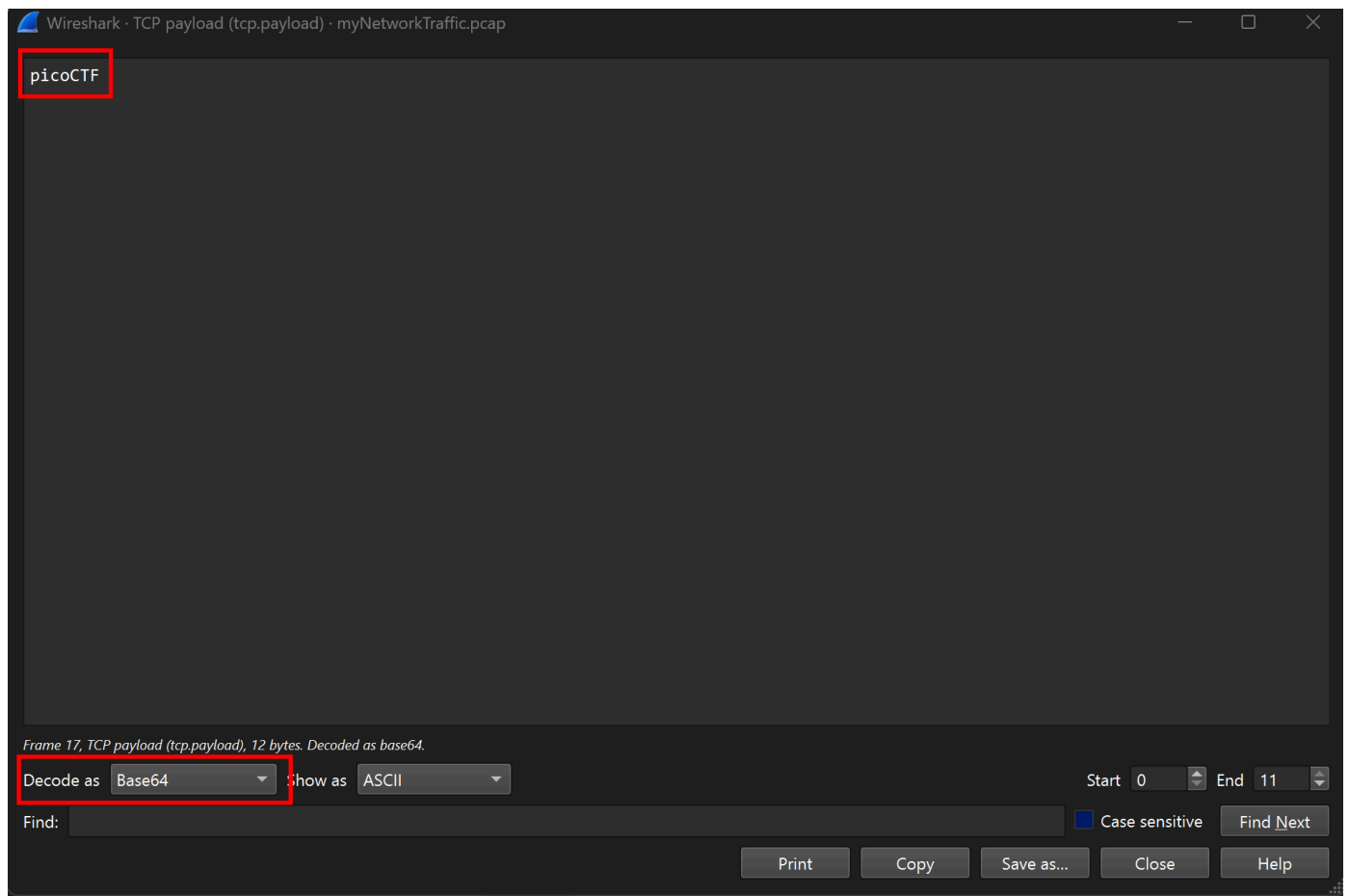
Packets with 12 length tcp payload has bytes that has == at the end which can be assumed as Base64 encryption. If I click show bytes on tcp payload of these packets, I can decode as Base64.

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11	0.002819	192.168.0.2	192.168.1.2	TCP	52	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=12
12	0.003036	192.168.0.2	192.168.1.2	TCP	44	[TCP Retransmission] 20 → 80 [SYN] Seq=0 Win=8192 Len=4

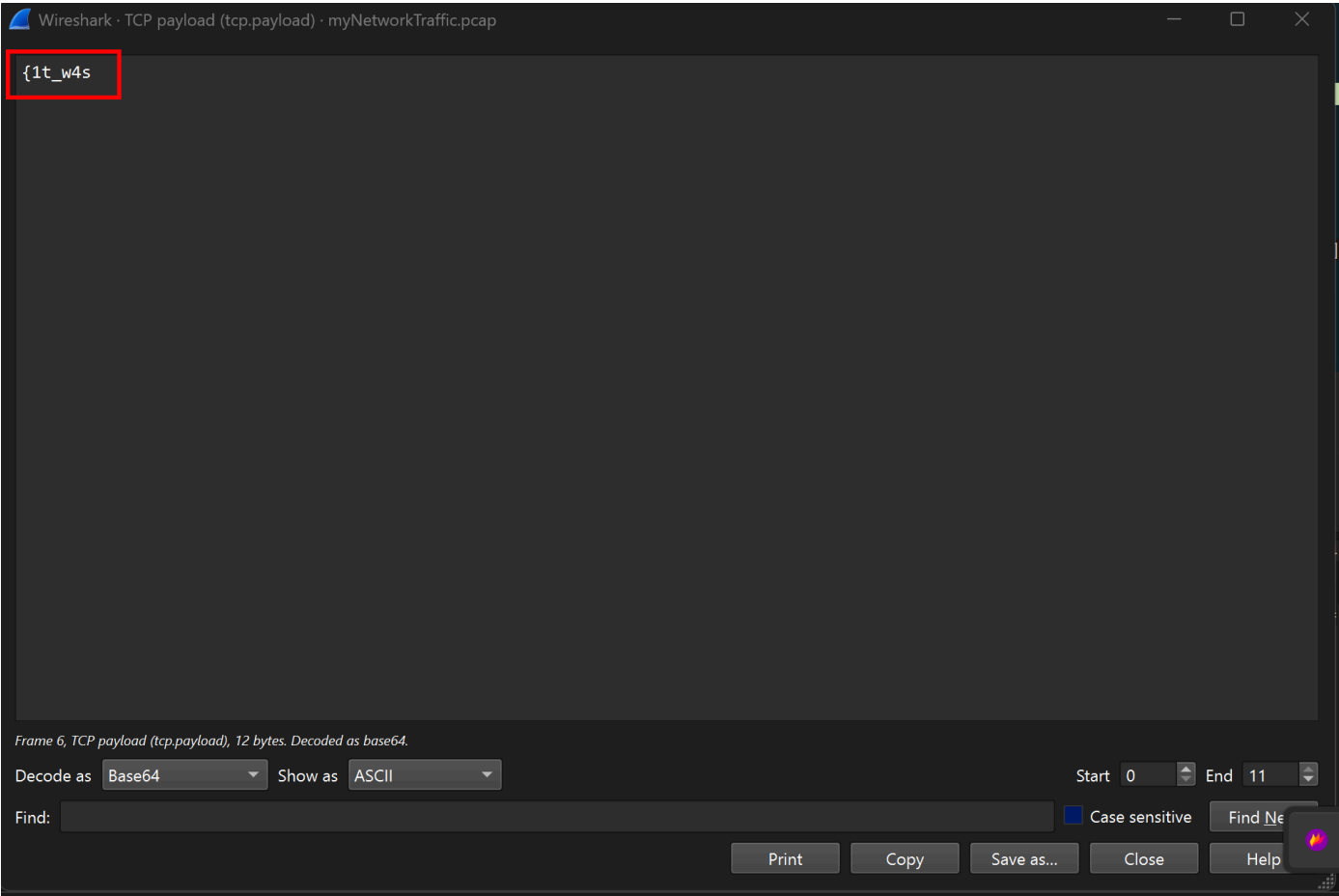
0101 = Header Length: 20 bytes (5)
Flags: 0x002 (SYN)
Window: 8192
[Calculated window size: 8192]
Checksum: 0xf4d1 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
[Client Contiguous Streams: 0]
[Server Contiguous Streams: 1]
TCP payload (12 bytes)
Retransmitted TCP segment data (12 bytes)

0000 45 00 00 34 00 01 00 00 40 06 f8 6e c0 a8 00 02 E..4....@..n...
0010 c0 a8 01 02 00 14 00 50 00 00 00 00 00 00 00 00P.....
0020 50 02 20 00 fd 41 00 00 63 47 6c 6a 62 30 4e 55 P...A...cgIjb0NU
0030 52 67 3d 3d Rg==

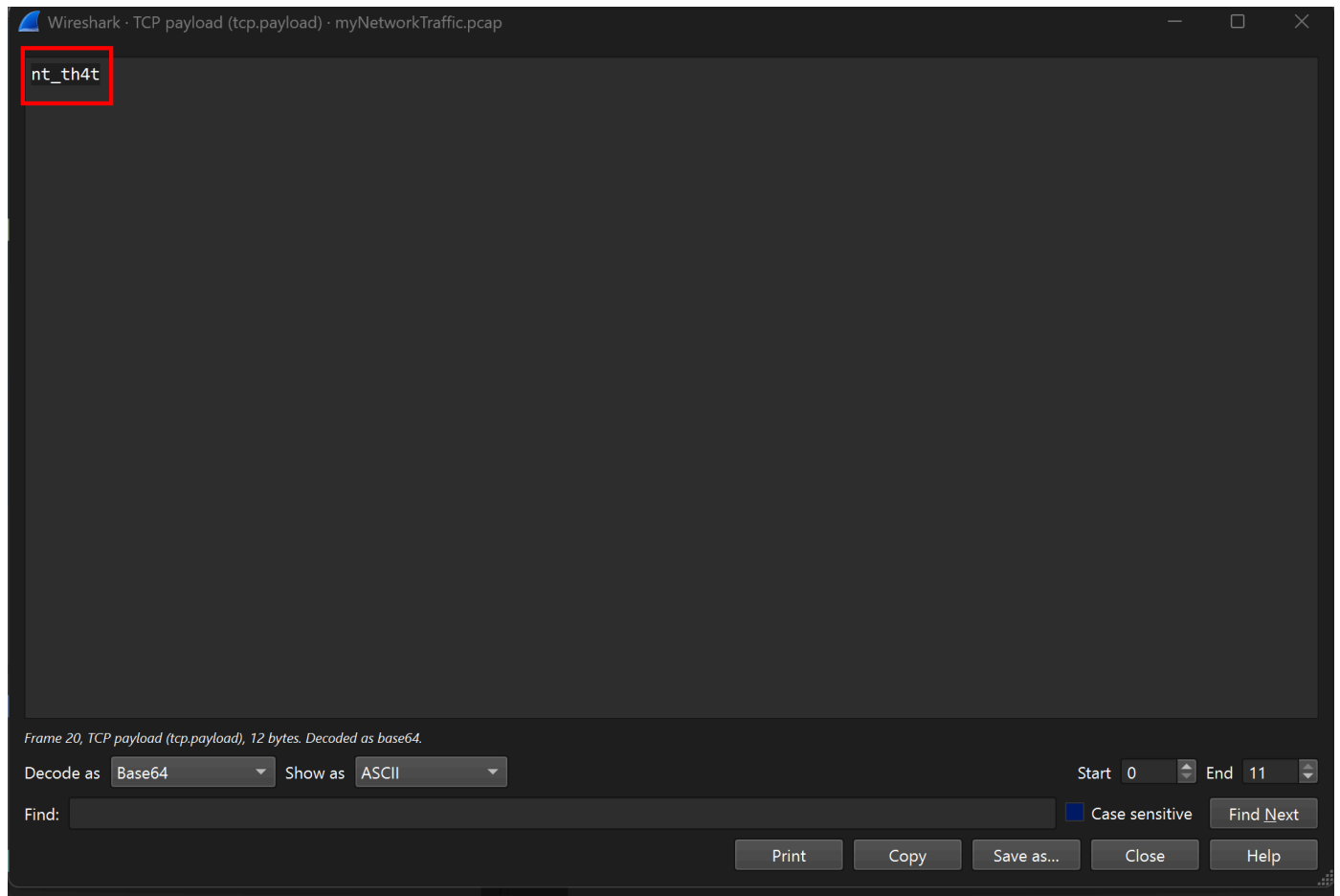
myNetworkTraffic.pcap Packets: 22 Profile: Default



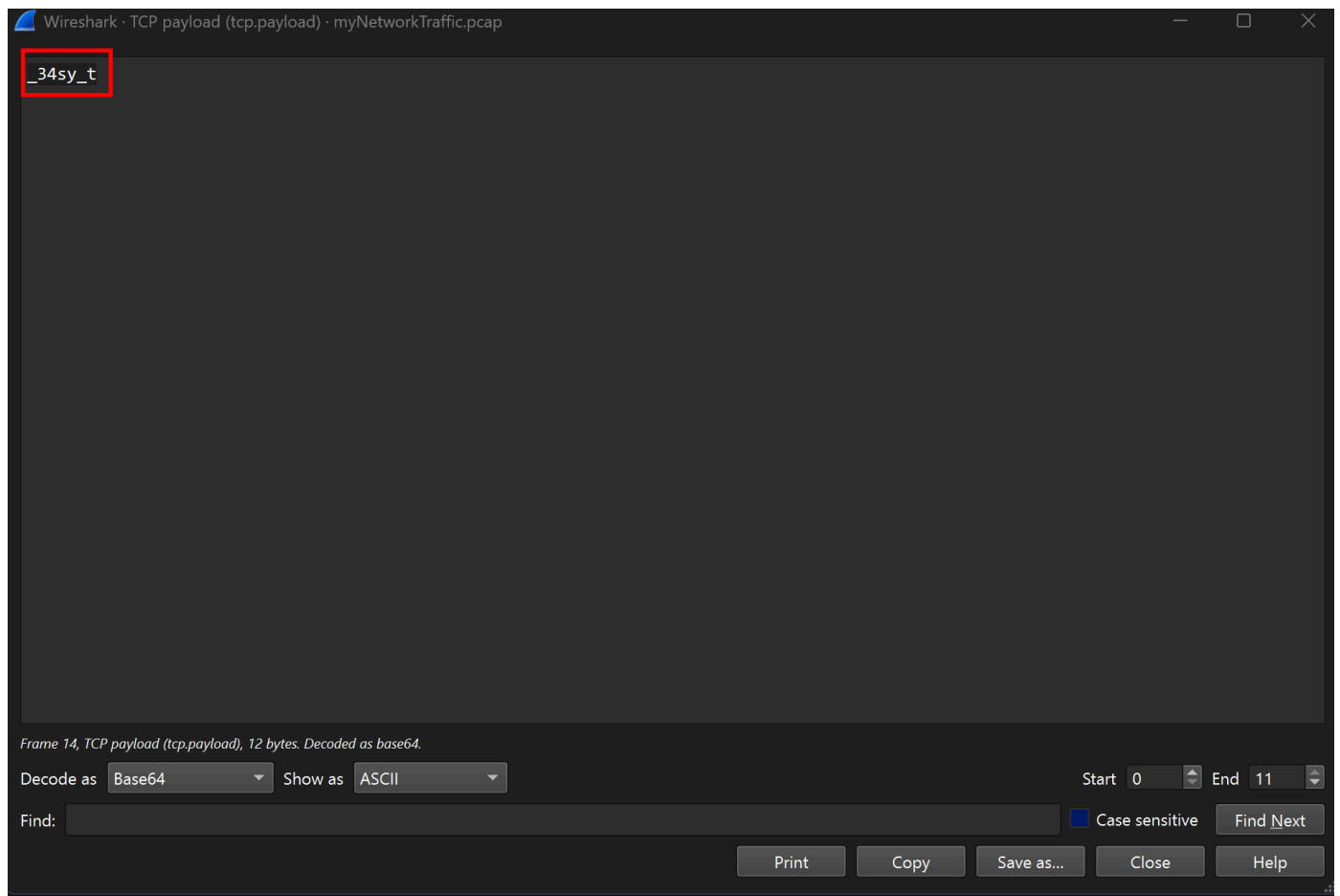
picoCTF



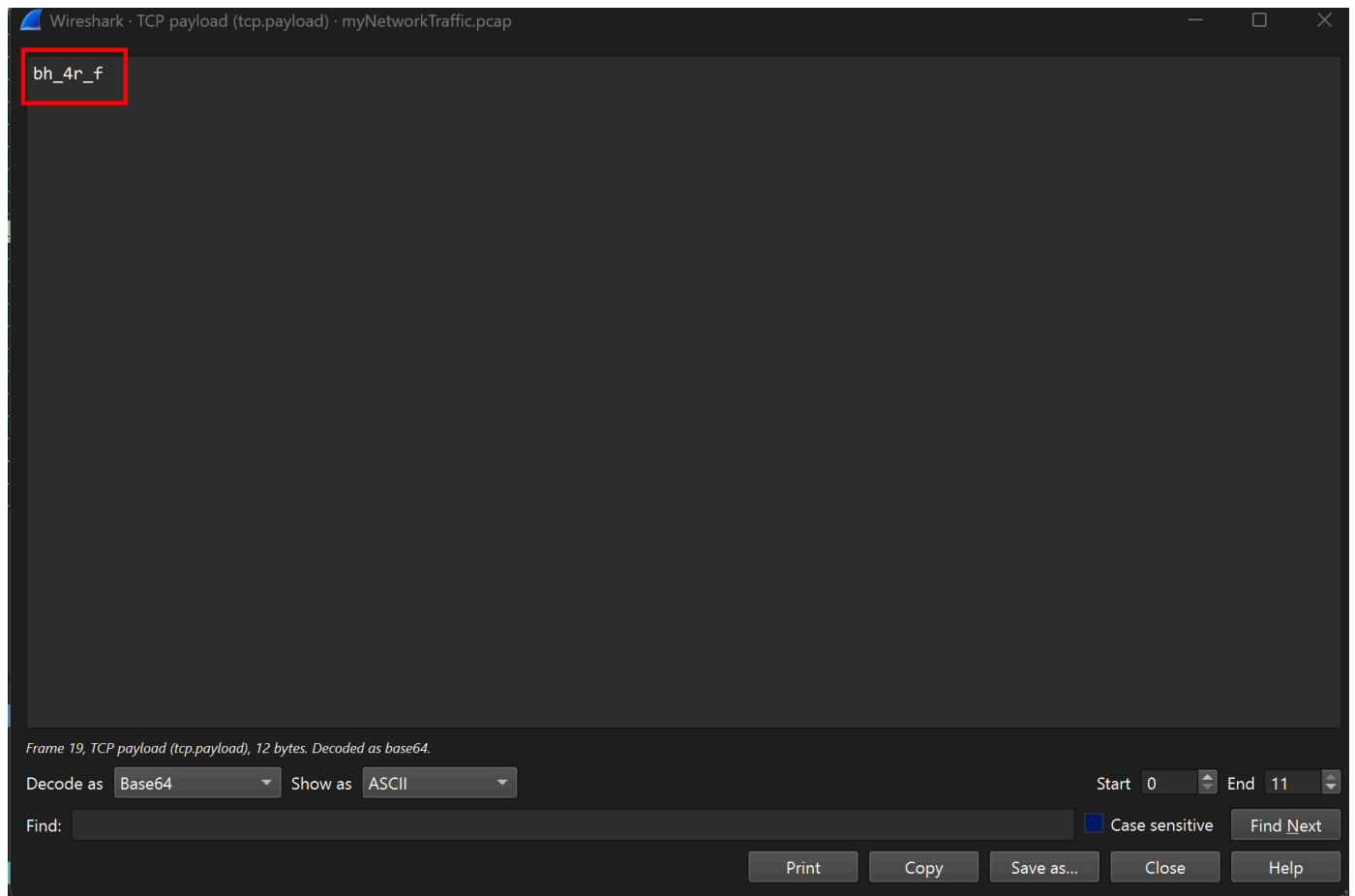
{1t_w4s



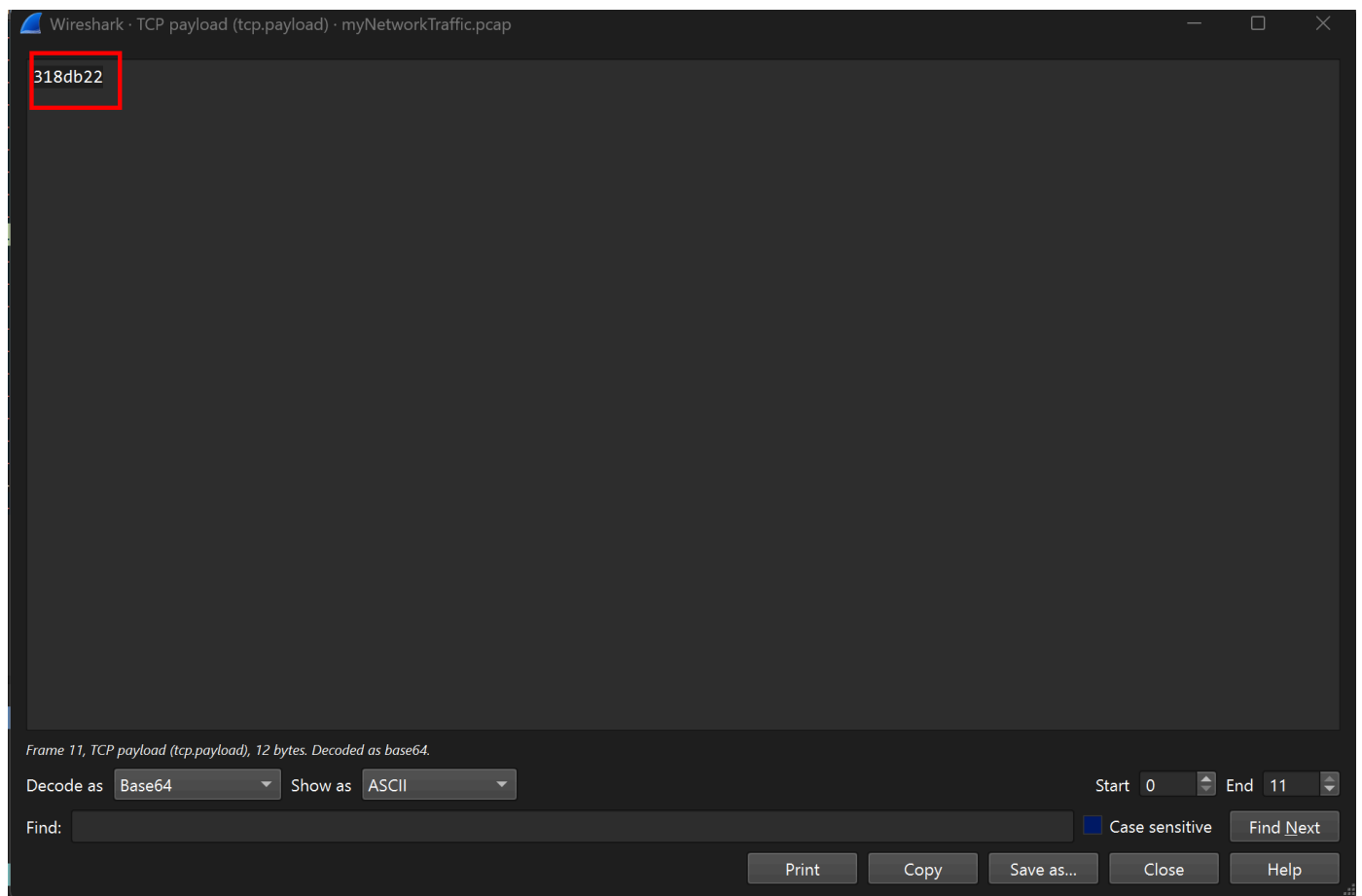
nt_th4t



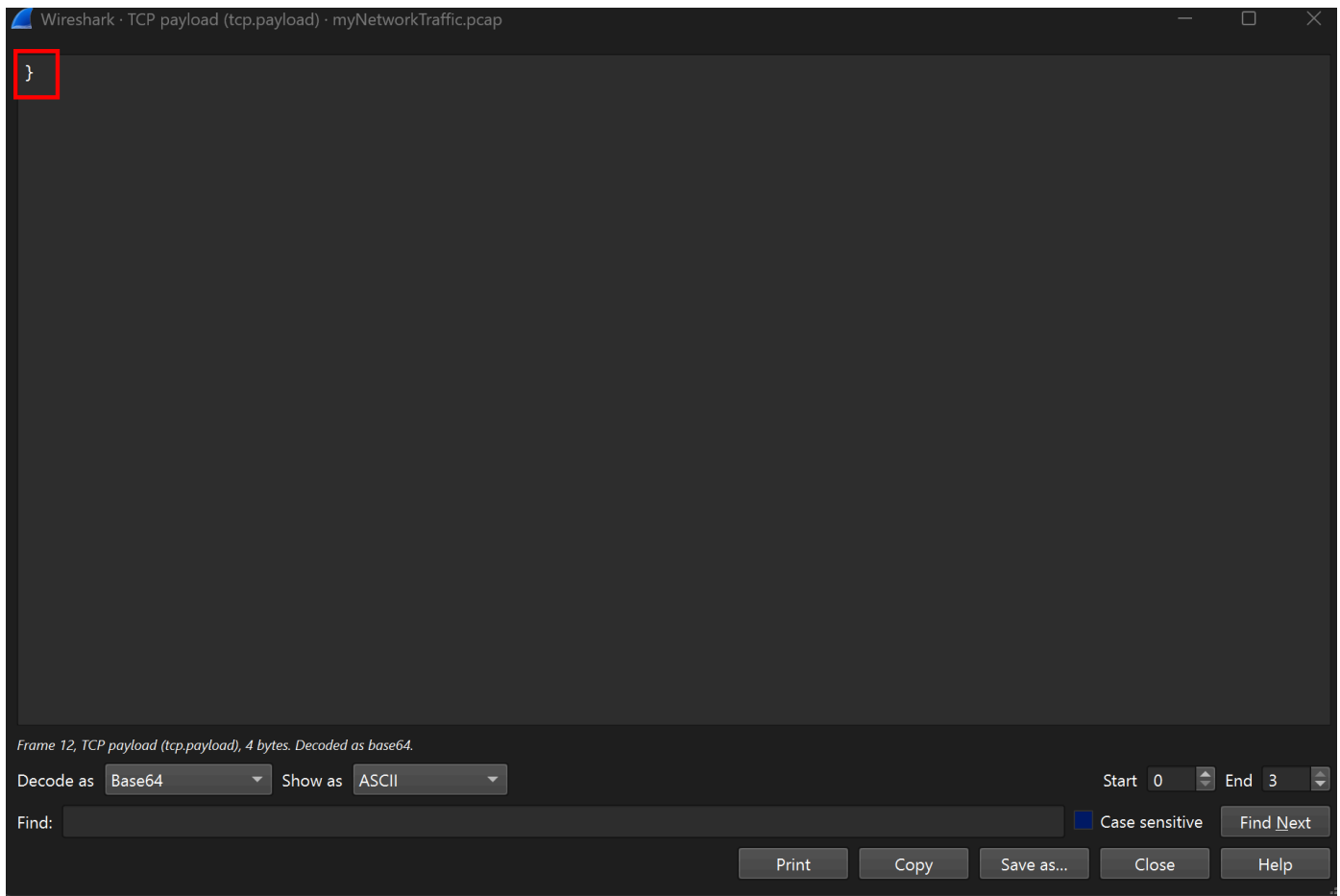
_34sy_t



bh_4r_f



318db22



}

Combining these will form the flag.

Quick Observation: Packets with len=12 and len=4 carried the flag

flag: picoCTF{1t_w4snt_th4t_34sy_tbh_4r_f318db22}